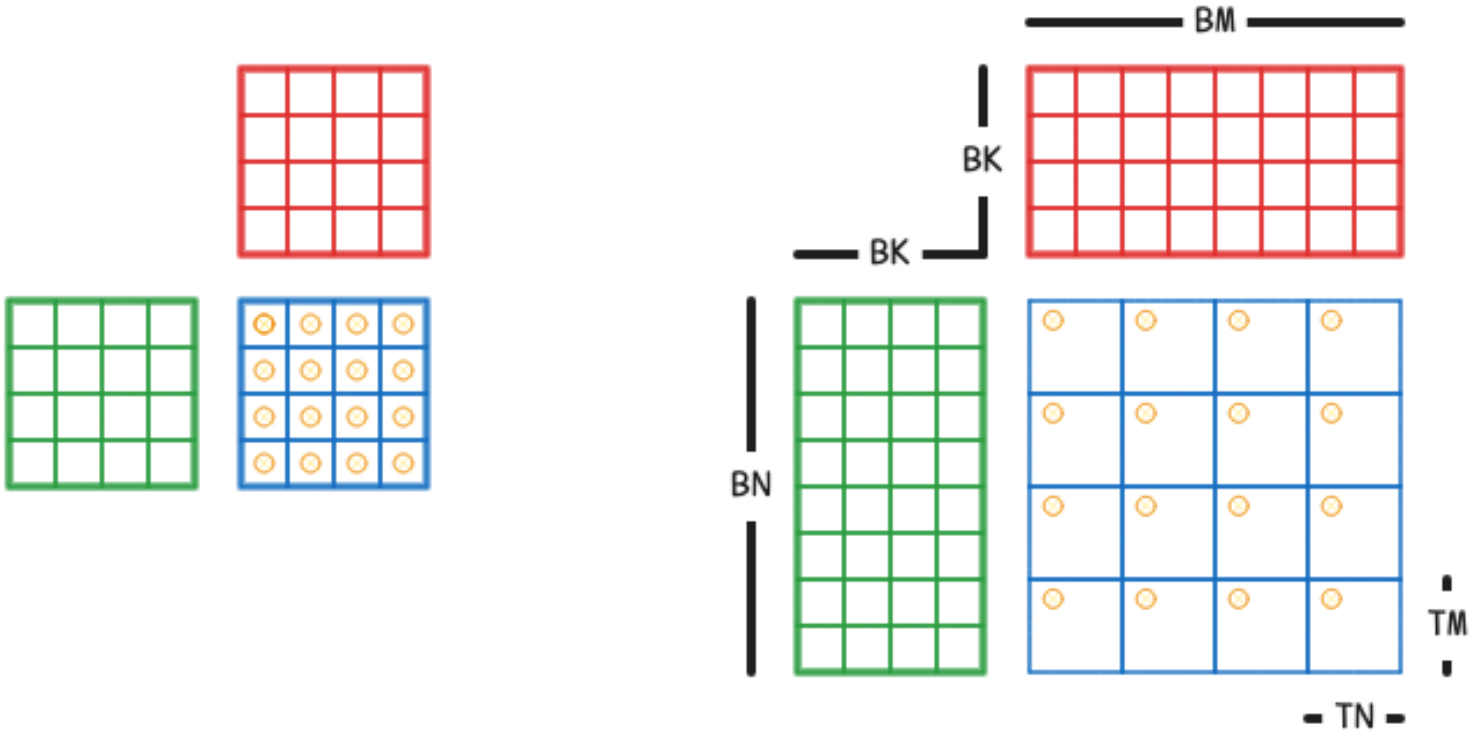
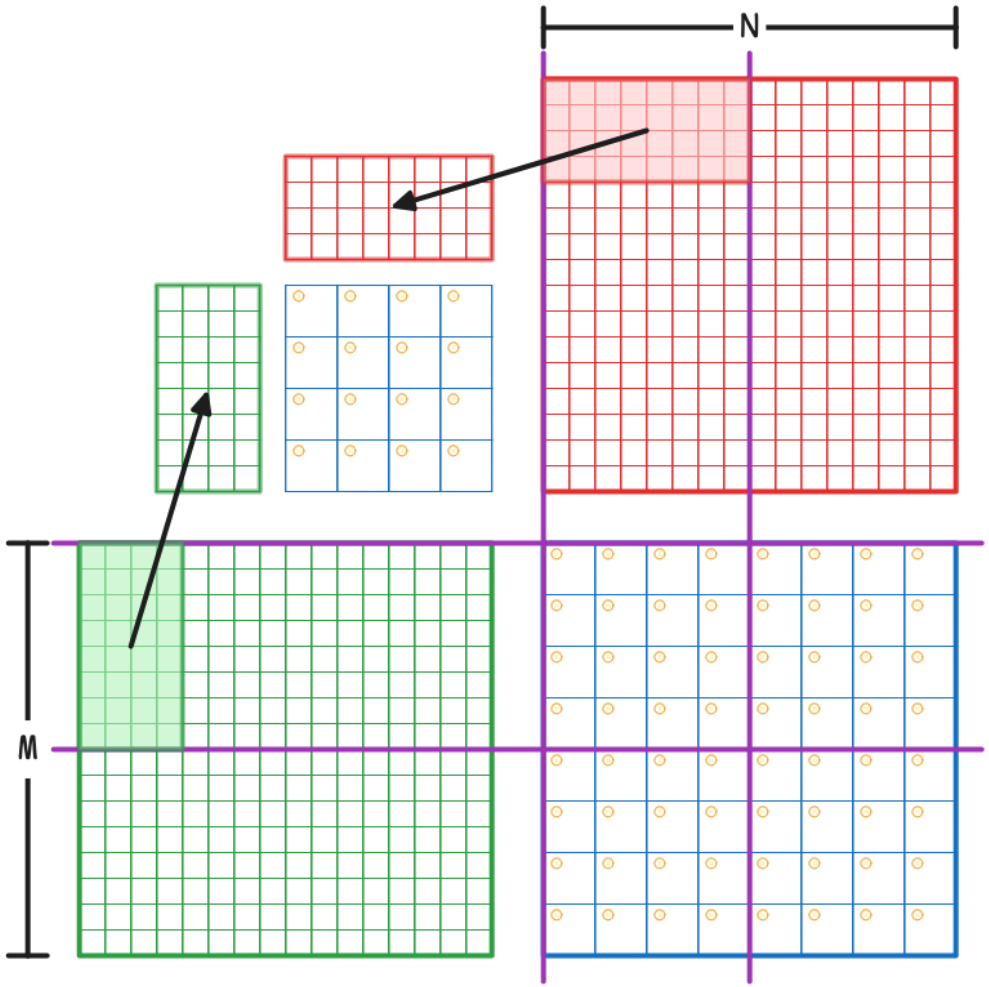
# Thread coarsening: Increasing AI

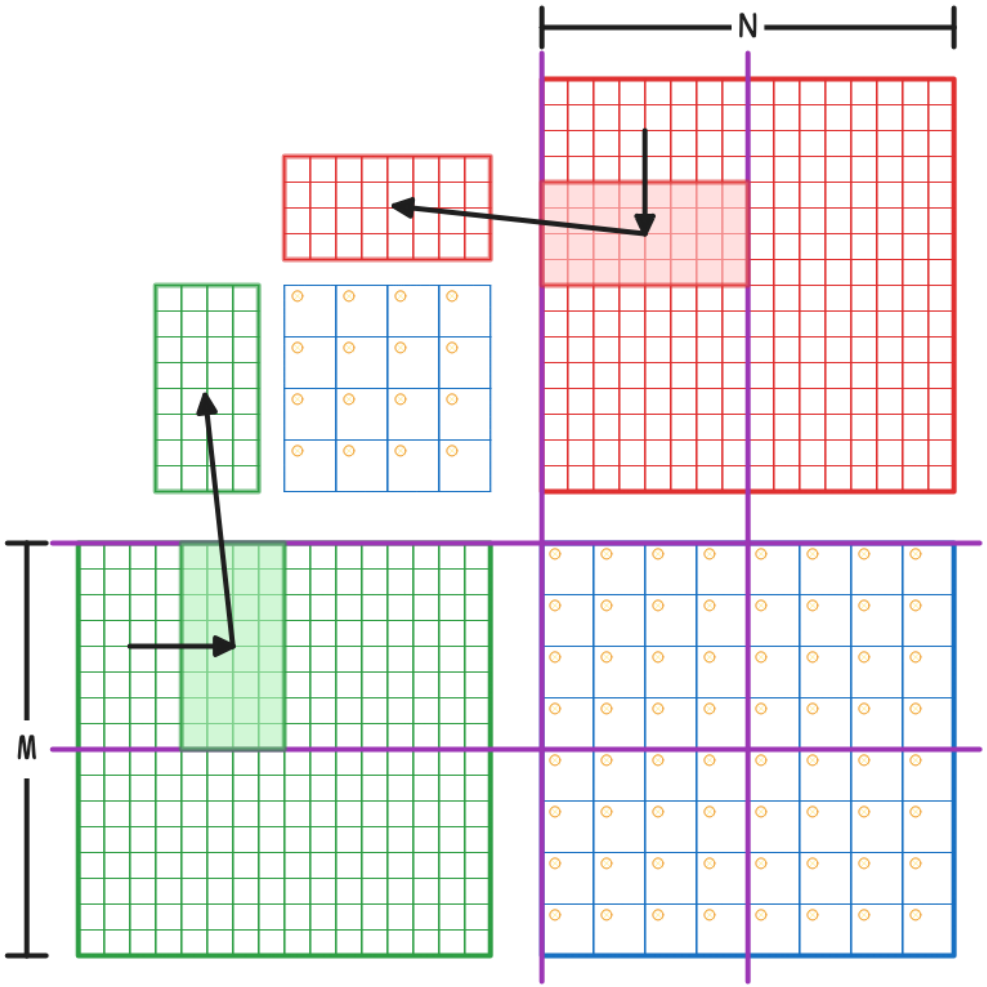
- Basic tiling assigns 1 element of the output to each thread.
- We can *coarsen* each thread, and assign a $TM \times TN$ portion of the output to each thread.
- We load larger chunks into SMEM, increasing data reuse and AI.
- One loaded $A$ value can update several columns; one loaded $B$ value can update several rows.



# Coarsened SGEMM visualization



# Coarsened SGEMM visualization



# Coarsening factors are a tradeoff

## Two levels of tiling

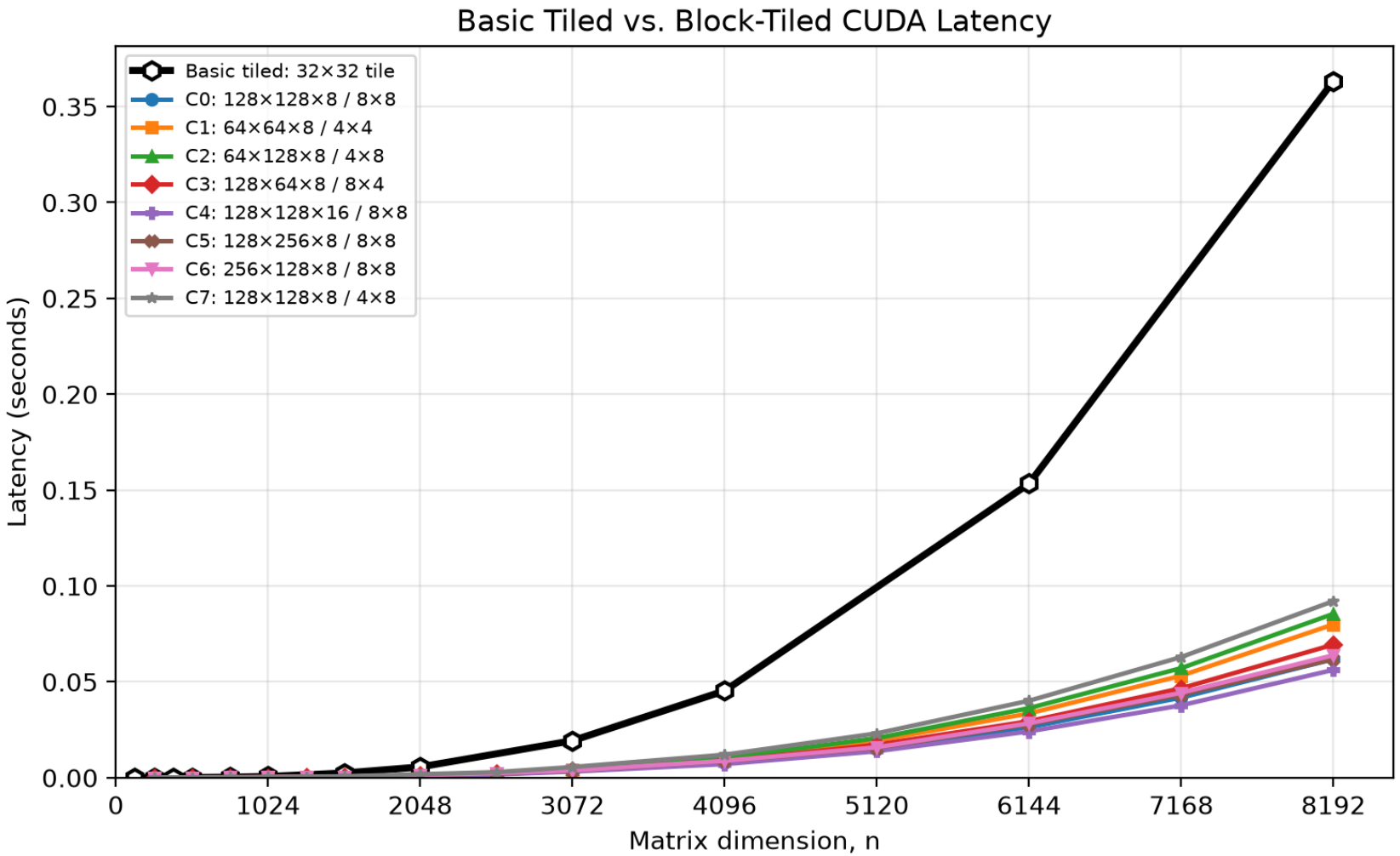
- A block computes a $BM \times BN$ output tile while advancing through $K$ in steps of $BK$.
- Each thread computes a $TM \times TN$ micro-tile in registers.
- Thread count: $\frac{BM}{TM}\frac{BN}{TN}$.

## Why not keep increasing them?

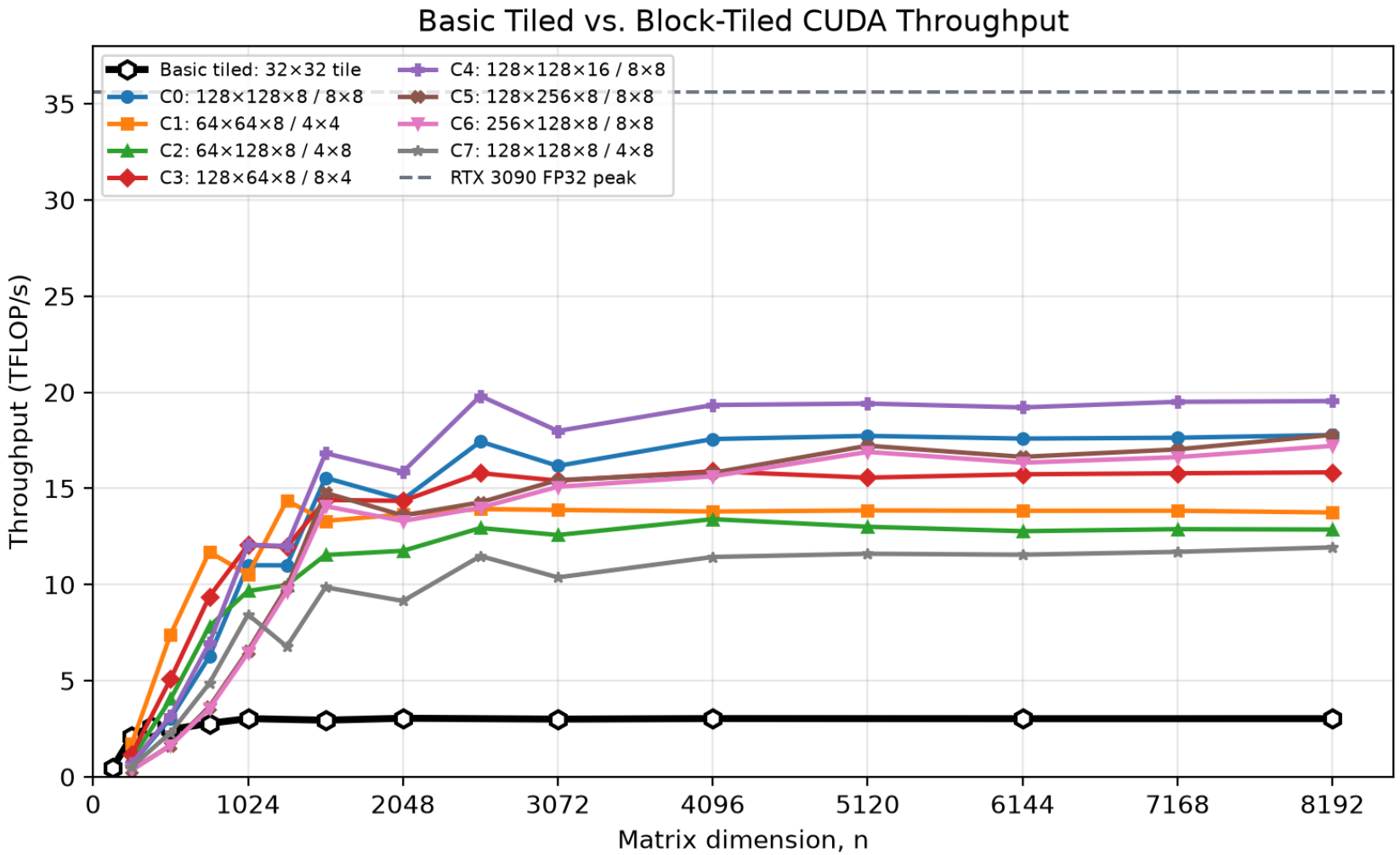
Larger tiles improve reuse and arithmetic intensity, but consume more registers and shared memory. Eventually occupancy falls and extra parallelism or reuse no longer helps.

| Config. | $BM \times BN \times BK$ | $TM \times TN$ | AI | Throughput |
|---|---|---|---|---|
| C1 | $64 \times 64 \times 8$ | $4 \times 4$ | 15.9 | 13.75 TFLOP/s |
| C4 | $128 \times 128 \times 16$ | $8 \times 8$ | 31.5 | **19.54 TFLOP/s** |
| C5 | $128 \times 256 \times 8$ | $8 \times 8$ | 41.8 | 17.79 TFLOP/s |

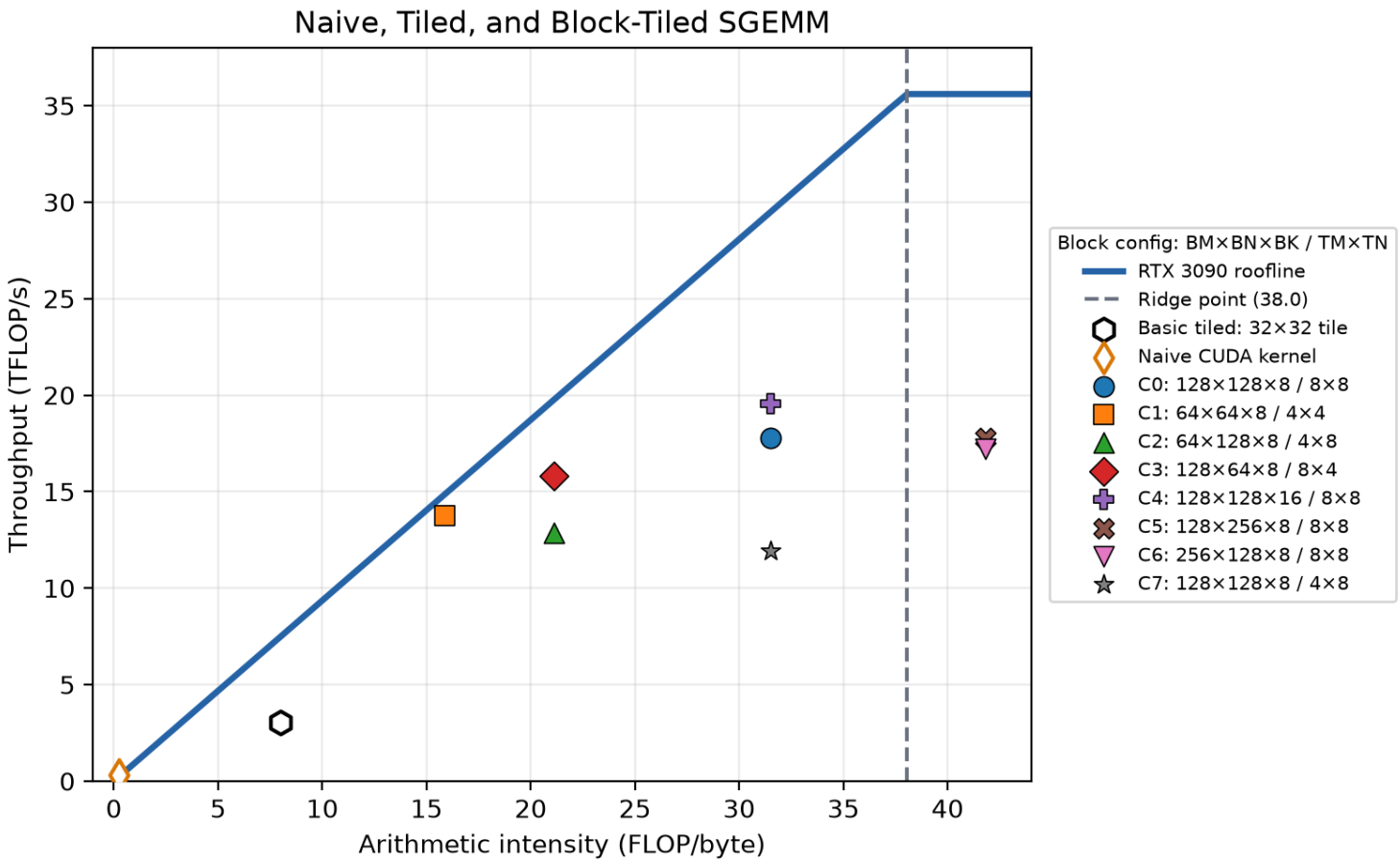
# Basic tiled vs. block-tiled latency



# Block tiling raises useful throughput



# Tiling configurations on the roofline




Reminder that the Roofline Model only tells us the upper bound!

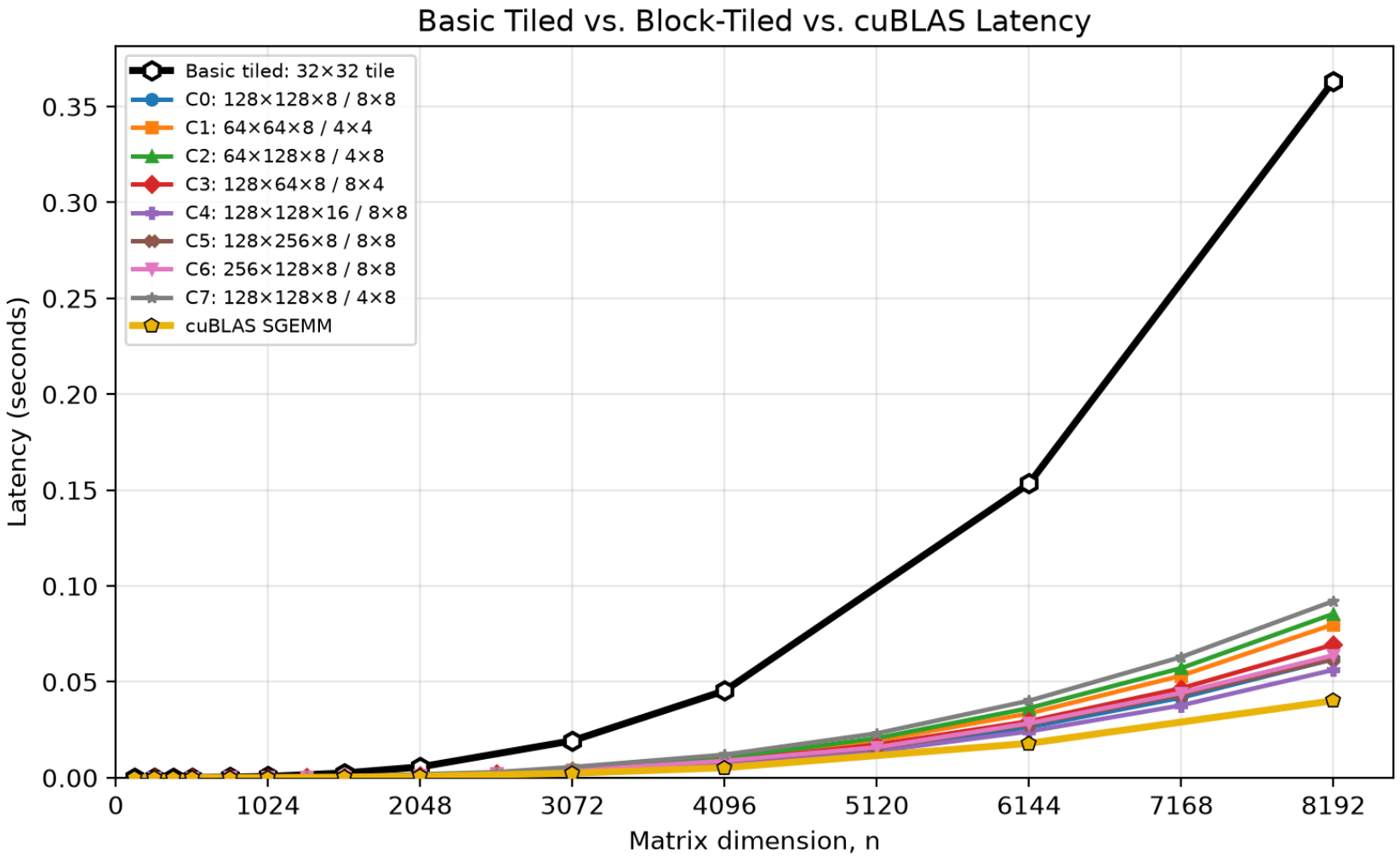
# We have only scratched the surface: cuBLAS

- cuBLAS is NVIDIA's architecture-tuned linear algebra library and the practical performance target for our SGEMM kernels.
- At $n = 8192$, cuBLAS reaches 27.38 TFLOP/s, while our fastest configuration (C4) reaches 19.54 TFLOP/s, or 71.4% of cuBLAS's throughput.
- cuBLAS completes the multiplication in 40.2 ms versus 56.3 ms for C4, about 1.40× faster.

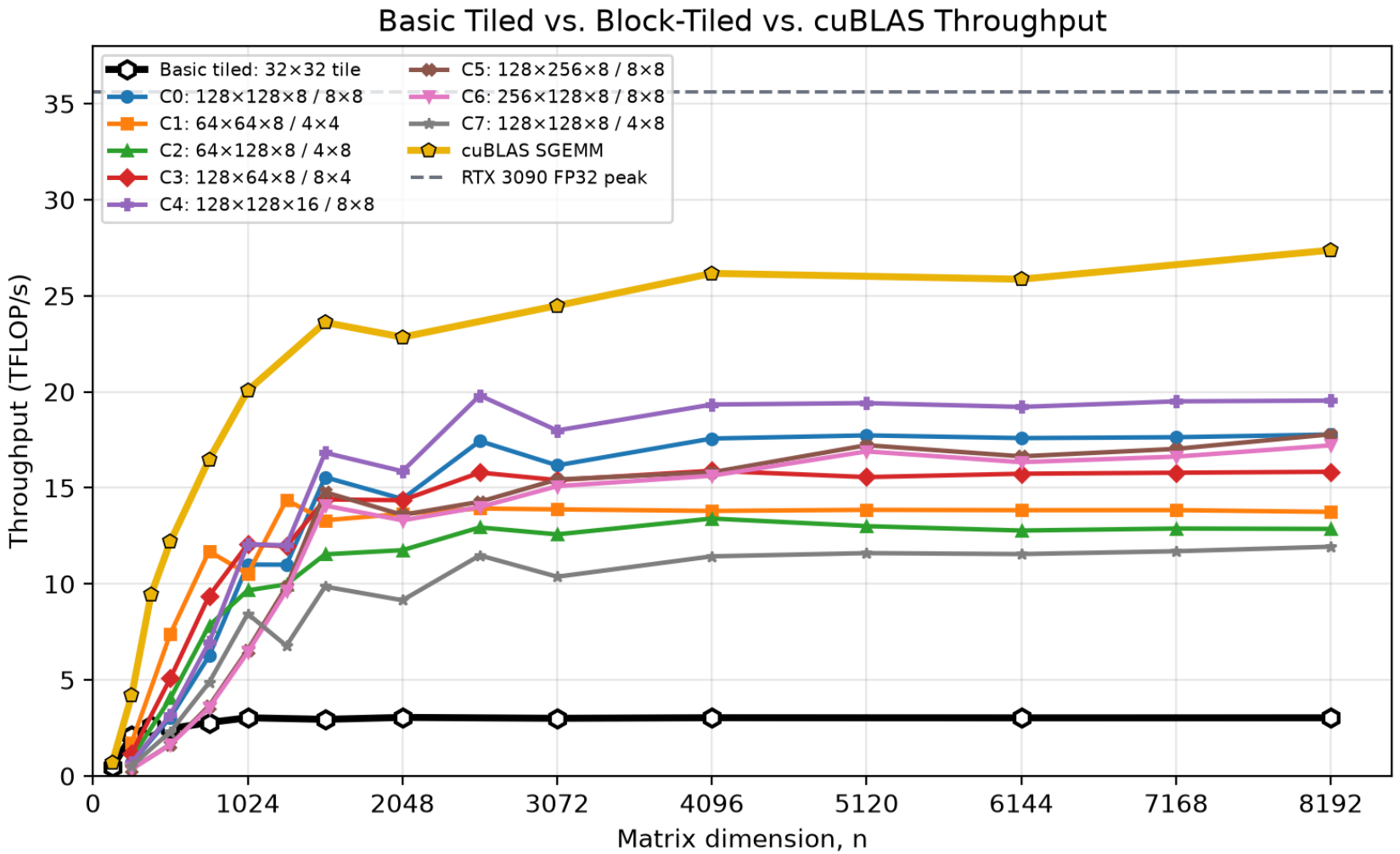
## The remaining gap

Production kernels combine deeper register and warp tiling, vectorized memory access, pipelined loads, and architecture-specific instruction scheduling.

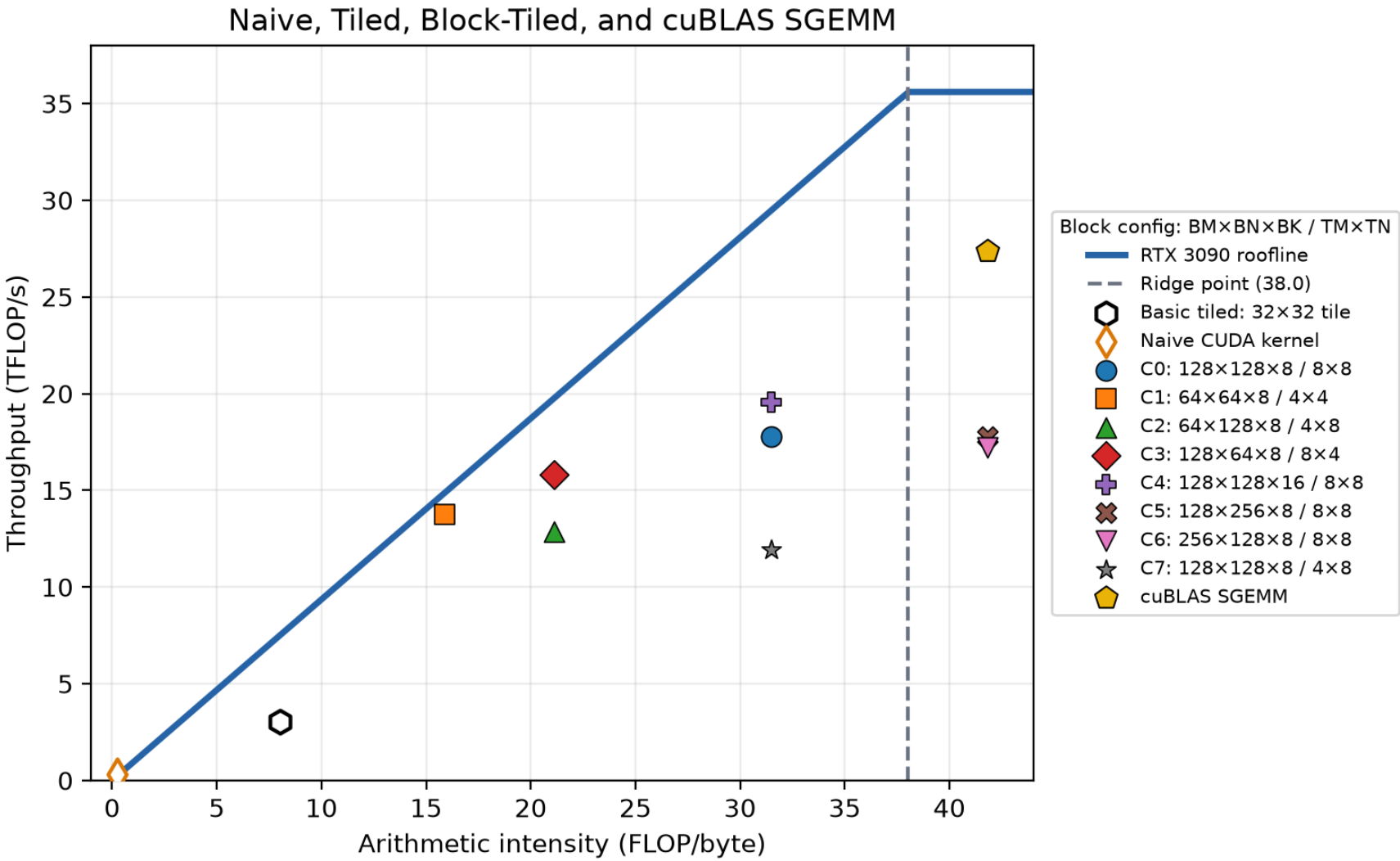
# Basic tiled vs. block-tiled latency vs. cuBLAS



# cuBLAS remains the performance target



# cuBLAS on the roofline




cuBLAS arithmetic intensity is estimated from the profiled $256 \times 128$ output-tile shape.

# We've come a long way

